Overnight Bitcoin Losses and the Timing of US Equity Repricing

Minwoo Song

Finance, Information Systems, and Economics
Lehman College, City University of New York, Bronx, NY, USA
minwoo.song@lehman.cuny.edu

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Abstract

We examine whether overnight Bitcoin losses are associated with lower US equity opening prices or predict subsequent intraday downside risk. Daily observations from January 2018 to October 2026 align hourly Bitcoin prices with US equity closures. A one-percentage-point Bitcoin log loss is associated with SPY and QQQ opening gaps lower by 0.108 and 0.125 percentage points. Conditional mean models show weak intraday return predictability. We extend the analysis using downside-probability models and conditional tenth-percentile regressions. Bitcoin losses are associated with a greater probability of a negative opening gap, but neither extension provides clear evidence of subsequent intraday downside predictability. Exchange and crisis-exclusion checks preserve the opening association, while retrospective forecasts show no intraday accuracy gain. Overlapping overnight windows prevent a causal or information-leadership interpretation. The contribution under investigation is the timing of equity repricing and the distinction between opening association and subsequent downside prediction; incremental information relative to futures remains unresolved.

Keywords: Bitcoin; overnight returns; downside risk; quantile regression; market integration.
JEL: G12; G14; G17.

1 Introduction

Bitcoin trades during US cash-equity market closures, making overnight cryptocurrency returns a possible indicator of information accumulated before the equity opening. However, a lower stock-market opening following a Bitcoin loss does not by itself imply that Bitcoin predicts returns available to an investor after observing that loss. Separating the opening adjustment from subsequent trading is therefore essential.

Existing research documents increasing crypto-equity interconnectedness (Iyer, 2022), high-frequency cross-asset spillovers (Leong and Kwok, 2025), and prediction of Monday equity returns from negative weekend cryptocurrency returns (Mourey et al., 2025). Gu et al. (2026) use overnight cryptocurrency returns to forecast VIX dynamics. Our exploratory analysis focuses on the location of the equity response within the daily return decomposition and its implications for forecasting. We do not claim that overnight spillovers or sentiment forecasting are new.

The economic distinction concerns both the average response and its distribution. An insignificant conditional mean coefficient can conceal a higher probability of negative returns or a deterioration in the lower tail. We therefore examine three margins: expected returns, the probability of a negative return, and the conditional tenth percentile. The probability model evaluates the frequency of losses; the quantile model evaluates the location of adverse outcomes without treating every observation as equally informative about the conditional mean.

2 Data and timing

The sample covers 2,202 NYSE sessions from January 2, 2018, through October 6, 2026. The final year is incomplete. We obtain Gemini BTC/USD hourly prices from CryptoDataDownload and use Bitstamp prices as an exchange robustness check. SPY and QQQ opening and closing prices are downloaded from Yahoo Finance; VIX and SKEW observations come from Cboe. The accompanying workbook contains the daily analysis panel, definitions, source locations and results. Frozen source files and construction code accompany the replication package.

For each equity session t, the Bitcoin signal starts at the preceding NYSE close and ends at 09:00 New York time. The previous close is normally 16:00, or 13:00 on an early-close session. Unix timestamps are converted consistently; daylight saving time, holidays and weekends are handled using the exchange calendar. Monday and post-holiday signals include the entire intervening closure. Calendar closure length enters the regressions.

$$b_t = 100 \log \left(\frac{P^{BTC}_{t,09:00}}{P^{BTC}_{\text{previous equity close}}} \right). \tag{1}$$

Endpoint prices are closes of hourly bars ending at the specified time. In particular, the 09:00 signal uses the bar ending immediately before 09:00, never the bar starting at 09:00. No interpolation or forward-filling is used. Primary-exchange endpoints are available for every sample session. The last 30 minutes before the cash-equity opening are deliberately excluded.

$$g_{it} = 100 \log(O_{it}/C_{i,t-1}), \qquad d_{it} = 100 \log(C_{it}/O_{it}). \tag{2}$$

Here g is the opening gap and d is the intraday return for ETF i. Their sum is the close-to-close log return. Raw ETF prices are used, so gaps include ex-dividend adjustments; a submission version

should account for distributions or use corresponding cash indices. VIX and SKEW outcomes are close-to-close index-point changes. They measure option-derived conditions, not returns on individual options.

The opening-gap and VIX-change windows overlap the Bitcoin measurement window. These outcomes establish associations with daily repricing, not realizable returns from a 09:00 trade. Intraday equity returns occur strictly after the signal and provide the cleaner predictive outcome. Futures and contract-level options have not been estimated in this version.

3 Empirical models

3.1 Conditional mean and asymmetric responses

$$L_t = \max(-b_t, 0), \qquad G_t = \max(b_t, 0). \tag{3}$$

$$y_t = \alpha + \beta_L L_t + \beta_G G_t + \gamma' Z_{t-1} + \delta' D_t + \varepsilon_t. \tag{4}$$

Z includes the prior SPY close-to-close return, prior VIX close and lagged 20-session SPY return standard deviation. D includes closure length and weekday indicators. SKEW regressions additionally control for prior SKEW. All baseline regressors are observable by the signal cutoff. Ordinary least squares uses Newey–West covariance estimates with five lags and a finite-sample covariance correction; reported p-values are asymptotic and two-sided. Missing controls reduce the principal sample to 2,201 observations. Equal signed-return sensitivity requires $\beta_L + \beta_G = 0$, which we test directly.

3.2 Downside probability

For each ETF and return window, define $I_{it} = \mathbf{1}\{r_{it} < 0\}$. We estimate a linear probability model with the same regressors:

$$I_{it} = a + \lambda_L L_t + \lambda_G G_t + c' Z_{t-1} + e' D_t + u_{it}. \tag{5}$$

The coefficient $100\lambda_L$ is the percentage-point change in the fitted probability of a negative return per one-percentage-point BTC log loss. HAC inference follows the mean model. A zero return is coded as nonnegative. This is a linear approximation, not a structural probability model. We report the number of fitted values outside $[0, 1]$ rather than clipping them. Benjamini–Hochberg adjustment is applied across the four loss-coefficient tests within this extension.

3.3 Conditional lower tail

Following the quantile-regression framework of Koenker and Bassett (1978), we estimate the conditional tenth percentile of intraday returns:

$$Q_{0.10}(d_{it} \mid X_t) = a_{0.10} + q_L L_t + q_G G_t + c'_{0.10} Z_{t-1} + e'_{0.10} D_t. \tag{6}$$

The estimator minimizes the sum of check losses $\rho_\tau(u) = u[\tau - \mathbf{1}\{u < 0\}]$. A negative q_L would indicate a lower conditional downside quantile after a BTC loss. This coefficient is neither expected shortfall nor a causal crash effect. We obtain 95% percentile confidence intervals using 499 circular moving-block pairs-bootstrap samples of five consecutive trading sessions, preserving the regressor–outcome pairing within each sampled block. The seed is 314159, the first six digits of π, fixed

before this rerun as an outcome-independent reproducibility convention. No alternative seeds were screened to obtain preferred results. These are pointwise intervals; bootstrap block-length sensitivity and multiplicity across the entire project remain unexamined. Both extensions were selected after inspection of the baseline results and are explicitly exploratory.

4 Results

4.1 Mean returns and opening repricing

Table 1 shows a negative association between overnight Bitcoin losses and equity opening gaps. A one-percentage-point Bitcoin log loss is associated with a 0.108-percentage-point lower SPY gap and a 0.125-percentage-point lower QQQ gap, holding controls and positive Bitcoin returns fixed. The corresponding intraday coefficients are -0.009 and -0.005, with p-values of 0.512 and 0.754. Their confidence intervals include economically modest effects in either direction; insignificance does not establish an exactly zero relationship.

Table 1: Overnight Bitcoin loss coefficients

Outcome	β_L	p value	95% interval	N
SPY gap	-0.1077	0.0001	[-0.1632, -0.0522]	2201
SPY intraday	-0.0091	0.5125	[-0.0364, 0.0182]	2201
QQQ gap	-0.1252	< 0.0001	[-0.1803, -0.0702]	2201
QQQ intraday	-0.0053	0.7535	[-0.0385, 0.0279]	2201
VIX change	0.2467	0.0029	[0.0844, 0.4091]	2201
SKEW change	-0.0472	0.3389	[-0.1439, 0.0495]	2195

Equity outcomes are percentage log returns; VIX and SKEW changes are index points. Slopes are per one percentage point BTC log loss. OLS with controls and five-lag HAC covariance.

The VIX-change coefficient is positive, 0.247 points per unit loss (p = 0.0029), while the SKEW coefficient is insignificant. Negative Bitcoin returns therefore coincide with greater option-implied volatility in this specification, without corresponding evidence for daily SKEW changes. This does not identify fear, demand for put protection, or a causal volatility shock.

The opening-gap coefficients on positive Bitcoin returns are 0.031 for SPY and 0.049 for QQQ. Tests of equal signed-return sensitivity yield p-values of 0.028 and 0.030, indicating greater downside sensitivity in these exploratory models. Benjamini–Hochberg adjustments across eight full-sample loss tests retain the opening-gap and VIX associations. This correction does not cover every specification explored in the project.

A descriptive event comparison identifies 211 sessions following a simple overnight Bitcoin loss of at least 3%. SPY averages a -0.453% opening gap and a +0.094% intraday return on these sessions, compared with +0.086% and +0.006% on other sessions. The event means are unadjusted and are not accompanied by a causal interpretation or a formal event-mean test.

4.2 Downside frequency and lower-tail returns

The probability estimates reinforce the distinction between opening and subsequent trading. A one-percentage-point overnight BTC log loss is associated with a 3.085-percentage-point increase in the probability of a negative SPY opening gap and a 3.469-point increase for QQQ (Table 2).

Both remain significant under within-family adjustment. The fitted intraday probability changes are 0.289 points for SPY ($p = 0.606$) and 0.025 points for QQQ ($p = 0.961$). Their intervals include zero and do not support reliable prediction of a negative intraday return. Five SPY-gap and seven QQQ-gap fitted probabilities fall outside $[0, 1]$; neither intraday model has such values. Accordingly, the gap slopes should be read as local linear associations, not globally valid probability changes.

Table 2: Downside probability models

Outcome	Loss slope (pp)	p value	95% HAC interval (pp)	N
SPY gap	3.085	< 0.0001	[1.940, 4.230]	2201
SPY intraday	0.289	0.6059	[-0.808, 1.385]	2201
QQQ gap	3.469	< 0.0001	[2.257, 4.681]	2201
QQQ intraday	0.025	0.9608	[-0.959, 1.008]	2201

The dependent variable equals one for a negative return. Slopes and intervals are probability percentage points per one percentage point of BTC log loss. Controls match the mean model. Five-lag HAC covariance. The opening-gap outcome overlaps the BTC signal.

The tenth-percentile estimates are negative but imprecise (Table 3). The SPY coefficient is -0.0171, with a 95% block-bootstrap interval of $[-0.0472, 0.0169]$; the QQQ coefficient is -0.0040, with interval $[-0.0446, 0.0378]$. Of 499 requested bootstrap fits, 499 were retained for SPY and 498 for QQQ; one QQQ fit reached the iteration limit and was excluded under the fixed convergence rule. A negative point estimate alone is insufficient evidence of lower-tail predictability. These intervals still permit modest adverse effects, so the results do not establish the absence of all downside information. The tail model is estimated in sample and has not been evaluated using out-of-sample quantile loss or coverage tests.

Table 3: Conditional tenth percentile of intraday returns

Outcome	Loss slope	95% bootstrap interval	N	Valid draws
SPY intraday	-0.0171	[-0.0472, 0.0169]	2201	499
QQQ intraday	-0.0040	[-0.0446, 0.0378]	2201	498

Slopes are changes in the conditional tenth percentile, in percentage log-return units, per one percentage point of BTC log loss. Pointwise 95% percentile intervals use 499 circular block-bootstrap draws, block length five sessions, seed 314159. They are not HAC confidence intervals and are not adjusted for multiple testing.

4.3 Stability and forecasting

The gap relationship is weak in 2018–2019 and stronger in subsequent subsamples. Excluding March 2020 preserves both the negative gap association and the positive VIX association. Bitstamp produces the same main qualitative pattern. Controlling for the equity opening gap in an at-open specification still yields insignificant Bitcoin-loss intraday coefficients. This conditional specification uses information available at the opening rather than at 09:00.

Notes: Subsample and exchange tests are exploratory. The positive 2023–2026 intraday estimate suggests reversal in that subsample; its p-value is not adjusted for the full set of robustness tests. Bitstamp has a shorter available sample.

We compare expanding-window intraday forecasts from the controls-only model with forecasts adding Bitcoin loss and gain terms. Evaluation starts in January 2023 and covers 943 sessions. Each

Table 4: Subsample sensitivity

Sample	Gap slope	p value	Intraday slope	p value
2018-2019	-0.0025	0.8512	-0.0123	0.4827
2020-2022	-0.1536	0.0005	-0.0174	0.4563
2023-2026	-0.1700	< 0.0001	0.0315	0.0401
Exclude March 2020	-0.0710	< 0.0001	0.0025	0.8211

SPY outcomes; same controls and inference as the baseline. Subsample tests are exploratory and not jointly adjusted for multiplicity.

forecast is estimated using strictly earlier observations. Relative to the baseline, out-of-sample R^2 is -0.101% for SPY and -0.194% for QQQ: the added Bitcoin variables slightly increase mean squared error. These are retrospective forecast simulations, not a prospectively reserved holdout, and no formal test of forecast-loss differences or trading profitability is claimed.

5 Interpretation and the futures comparison

The proposed next stage asks whether Bitcoin adds information beyond synchronized S&P 500 and Nasdaq-100 futures. CME equity futures trade during much of the cash-market closure, so cash-market opening gaps cannot isolate information originating in cryptocurrency markets. Gu et al. (2026) already provide S&P 500 futures data in their replication archive; their exact control specification must be checked before any novelty claim is made.

A daily specification would measure Bitcoin and ES/NQ futures over the same prior-close-to-09:00 interval, then predict strictly subsequent futures returns, equity intraday returns or intraday downside risk. The baseline should include signed futures movements and overnight futures volatility to make the incremental comparison demanding. Futures prices must refer to consistent contracts within each return interval, with an explicit roll rule and verified timestamps. Full-day futures settlement returns cannot substitute for the synchronized premarket controls.

$$y_t^{\text{post}} = \alpha + \beta_L L_t + \beta_G G_t + \theta' F_t^{\text{pre}} + \gamma' Z_{t-1} + \delta' D_t + \varepsilon_t. \tag{7}$$

Equation (7) is a proposed design, not an estimated result. Reverse-direction lead–lag tests and independently identified macro and crypto-origin events would help distinguish shared news, information leadership and transmission. A residual Bitcoin return is not automatically a structural crypto shock. Likewise, attenuation after controlling for futures could reflect information already transmitted through futures, rather than proving that shared news explains everything.

The new models broaden the outcome examined, but additional specifications do not by themselves establish novelty. The substantive interpretation depends on synchronized price information and defensible shock classification. In particular, lower opening prices, a higher likelihood of opening losses and higher daily VIX changes can all reflect the same accumulated overnight news. A future study should distinguish this explanation from crypto-origin transmission and should evaluate forecast additions against an information set available at a precisely stated decision time.

6 Conclusion

Overnight Bitcoin losses are associated with lower US equity opening prices, while broad-sample evidence of subsequent intraday declines is weak. The timing distinction limits the predictive interpretation of overnight associations. The present evidence motivates a synchronized futures comparison but does not establish incremental information, a fear mechanism or causality. A publication contribution depends on resolving those questions rather than extending the sample alone.

The downside-probability and lower-tail extensions test whether mean regressions miss a different form of intraday risk. Their estimates and uncertainty intervals should be interpreted jointly rather than selecting the most favorable specification. The current study offers an empirical basis for refining the timing question, not a completed causal or futures-based contribution.

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Conflict of Interest

The author declares no conflict of interest.

Data Availability Statement

The data used in this study are available from the author upon request.